

Nicholas Hargreaves, PMP

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Professional Summary

High-impact Sr. Principal Software Engineer and Technical Program Management (TPM) Leader with a distinguished record of executing complex, multi-million dollar cloud and embedded software integrations across global enterprise environments. Expert in architecting scalable connected-vehicle platforms, telemetry services, and high-throughput data pipelines. Proven success navigating major corporate transitions, managing matrixed international teams, and anchoring zero-downtime operations for millions of active endpoints.

Core Competencies

- **Technical and Strategic Leadership:** Enterprise Platform Architecture, Global Team Management (50+ Engineers), Cross-Functional Stakeholder Alignment, Vendor and 3rd Party Management, Budgetary Oversight (\$12M+), Strategic Risk Mitigation.
 - **Systems Architecture and Data Engineering:** Cloud Computing (Azure, AWS, GCP), Microservices, Platform-as-a-Service (PaaS), Distributed Systems, High-Throughput Event Streaming (Kafka, MQTTv5), ClickHouse, Containerization (Docker, Kubernetes/K8s).
 - **Software Engineering and Languages:** Node.js, Electron, Python, C#, C++, C, Rust, SQL, JavaScript, React, JSON, REST APIs, OpenSSL.
 - **Frameworks and Methodologies:** Agile/Scrum (Certified Scrum Product Owner), Digital Six Sigma (Green Belt), Software FMEA, CI/CD DevOps Pipelines, Jira/Azure DevOps, Enterprise Production SLA Compliance.
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Professional Experience

CARIAD, Inc. (Volkswagen Group) | Redmond, WA

Sr. Principal Software Engineer | Manager, Program Management | 2021 – Present

- **Ensured Production SLA Compliance** for live global MEB and PPE vehicle platforms across the EU and North America, closing the year with **zero open P1/P2 critical production tickets** and preventing platform-induced blockers[cite: 3].
 - **Architected Big Data Health Monitoring Infrastructure:** Migrated a legacy Python script into a production-grade, highly scalable Node.js service backed by ClickHouse and deployed on a Kubernetes (K8s) cluster; enabled scheduled cron-based scanning of **>1M active vehicles per run** tracking **>80 operational attributes**[cite: 3].
 - **Engineered High-Throughput Messaging Pipelines:** Refined GDC message processing workflows to handle production adoption loads exceeding **1,000 messages per second with zero dropped messages**, supporting features like Factory Reset and Digital Twin synchronization[cite: 3].
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- **Applied Statistical Rigor to Incident Resolution:** Led a data-driven investigation into EV Services command acknowledgment failures using production observability tools and MQTT logs; performed statistical analysis ($p < 10^{-2}$ large effect size) comparing EU vs. NA datasets to isolate vehicle session lifecycles as the true root cause, eliminating unnecessary platform changes[cite: 3].
- **Developed Cross-Platform Diagnostics Tooling:** Extended core backend APIs into an accessible, secure, zero-dependency **Electron desktop application** integrated with read-only GTC tenant authentication and compiled into a signed Windows installer for field engineer self-service[cite: 3].
- **Accelerated Global Customer Adoption:** Managed environment readiness and provided critical, on-site/virtual launch engineering support across international manufacturing hubs including Ingolstadt (Europe), Beijing (China), and Chattanooga (North America)[cite: 3].
- **Spearheaded Unified Data Environment (UDE) Initiatives:** Designed notification interfaces for VW.AC data integration and engineered a specialized pre-installation utility to preload large-scale vehicle datasets into UDE Kafka pipelines[cite: 3].

Kymeta Corporation, Inc. | Redmond, WA

Director, Program Management | 2016 – 2021

- **Spearheaded the program management office (PMO)** reporting to the C-Suite (COO, CFO, President) to deliver complex land-mobile satellite communications-on-the-move platforms[cite: 2].
- **Controlled an annual program portfolio budget exceeding \$12M**, driving hardware New Product Introduction (NPI), embedded software, and cloud-based OSS/BSS software solutions from quote-to-cash[cite: 2].
- **Established standardized agile tools and delivery processes** that synchronized multiple scrum teams, compressing cycle times and keeping milestones strictly on schedule[cite: 2].
- **Managed high-stakes technical interfaces** between commercial business units, core systems engineering, and strategic partners/customers including SpaceX, OneWeb, and Jaguar Land Rover[cite: 2].

Nuance Communications Inc. | Seattle, WA

Manager, Program Management | Principal Program Manager | 2013 – 2016

- **Built and mentored a senior team** of program managers delivering connected vehicle cloud services to major global automotive OEMs[cite: 2].
- **Owned delivery of head unit and cloud software integrations** for the high-profile Fiat Uconnect In-Vehicle Infotainment (IVI) system, acting as Product Owner within Jira-based Scrum frameworks[cite: 2].
- **Successfully managed aggressive timelines and budgets**, reporting status directly to Senior Accenture management on-site in Turin, Italy[cite: 2].
- **Designed and verified complex backend cloud integrations**, encompassing custom authentication schemes, secure VPN network architectures, and robust end-to-end network security layers[cite: 2].

Tweddle Group Inc. | Bellevue, WA

Senior Program Manager | 2012 – 2013

- **Directed the cross-functional delivery** of head unit applications and connected web services for the Chrysler Uconnect IVI platform, aligning milestones across Sprint and FCA/Chrysler executives[cite: 2].
- **Oversaw a multi-tiered client-server architecture** utilizing Java/Tomcat web servers and Oracle Enterprise databases to manage real-time vehicle provisioning, metrics reporting, and customer billing[cite: 2].
- **Guided UX/UI design, API development, and secure token authorization** for nested head unit software[cite: 2].
- **Negotiated and integrated 3rd-party content feeds** from major external digital vendors including Bing, Yelp, Pandora, and Slacker Radio[cite: 2].

BSQUARE Corp. | Bellevue, WA

Senior Program and Project Manager (Contracted to Ford-Microsoft) | 2006 – 2011

- **Managed a \$13M+ resource and operational budget** for the engineering and quality assurance of middleware applications within the iconic Ford MyFord Touch IVI system[cite: 2].
- **Integrated highly specialized 3rd-party vendor stacks** from Freescale, Audible, TeleNav, Nuance, Broadcom, Gracenote, and Airbiquity under immense delivery pressure[cite: 2].
- **Enforced strict Security Development Lifecycle (SDL)** and systems architecture audits across local and remote development arms[cite: 2].
- **Applied extensive knowledge of vehicle CAN bus networks** to ensure seamless interface compatibility with Bluetooth, USB, iOS, and Android mobile ecosystems[cite: 2].

Senior Program and Project Manager (Professional Services)

- **Owned delivery of Windows CE and Windows Mobile projects** for elite silicon and hardware vendors including Freescale, Dell, Palm, NXP, and Texas Instruments[cite: 2].
- **Drafted and secured formal commercial documents**, including highly technical RFQs, RFPs, and legal Statements of Work (SOWs) for budgets exceeding \$1M[cite: 2].

Motorola Ltd | Swindon, UK

Project Manager / Senior Engineer / Graduate Engineer | 1994 – 2006

- **Program Leadership (2001-2006):** Managed engineering teams (up to 17 heads across the UK and Poland) building UMTS/CDMA 1X Base Station and GSM Base Station Controller firmware[cite: 2]. Oversaw staffing budgets up to \$6M and achieved a perfect record with zero non-compliances during formal ISO TL9000 audits[cite: 2].
- **Process Automation:** Authored organizational software policies on automated code generation and utilized Digital Six Sigma methodologies to unlock a 10% reduction in cycle-time via C++ optimizations[cite: 2].
- **Systems and Firmware Engineering (1994-2001):** Designed and implemented POSIX-compliant, IP-based messaging schemes for shared infrastructure code[cite: 2]. Led embedded software engineering for GSM BTS Picocellular products from initial equalizing algorithms through to first office application (FOA) field deployments across Sweden, Germany, and the UK[cite: 2].

Professional Certifications and Affiliations

- **Chartered Engineer (CEng)** – Institute of Engineering and Technology (IET), UK[cite: 2]
 - **Project Management Professional (PMP)** – Project Management Institute (PMI)[cite: 2]
 - **Certified Scrum Product Owner (CSPO)** – Scrum Alliance[cite: 2]
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Education and Training

- **B.Eng. (Hons) Dip. Eng in Electronic and Communications Engineering** – University of Hull, UK[cite: 2]
- **Advanced Technical Certifications:** Software Failure Mode Effect Analysis (FMEA), Digital Six Sigma and Green Belt Training, Design for Six Sigma (Voice of Customer and Critical Parameter Management)[cite: 2]
- **Awards:** Kymeta Presidents' Award for Excellence in Program Delivery[cite: 2]